

Wenbo Ji

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Homepage Google Scholar GitHub

Education

Technical University of Munich

M.Sc. Electrical Engineering and Information Technology

Double degree with Tongji University

Oct 2023 – Nov 2026 Expected

Current Grade: 1.7

German grading scale: 1.0 is the best grade

Tongji University

M.Sc. Computer Science

Sept 2021 – Aug 2025

GPA: 4.5/5.0

Nanjing Tech University

B.Sc. Information and Computing Science

Sept 2017 – June 2021

GPA: 3.5/4.0

Research Interests

3D/4D Scene Representation Human-Centric Video Generation Embodied World Models

Selected Publications

* *Equal contribution* † *Corresponding author*

ViDS: Video Diffusion Shader using 3D Face Tracking

2026

Wenbo Ji, Davide Davoli, Zhe Chen, Liam Schoneveld, Matthias Nießner, Jiapeng Tang†

arXiv preprint

3D face tracking-conditioned video diffusion for expressive, identity-preserving portrait animation from a single image.

CSG-Fusion: Consistent Sparse-View Gaussian Splatting via Matching-based Fusion

2025

Yan Xia*†, Wenbo Ji*, Weirong Chen, Daniel Cremers

ICCV 2025 Workshop on E2E3D – Best Paper Award

Matching-based fusion of sparse-view pointmaps into compact, cross-view-consistent 3D Gaussians.

LiteTracker: Leveraging Temporal Causality for Accurate Low-latency Tissue Tracking

2025

Mert Asim Karaoglu, Wenbo Ji, Ahmed Abbas, Nassir Navab, Benjamin Busam, Alexander Ladikos†

MICCAI 2025

Causal temporal feature reuse with prior-motion initialization for accurate, low-latency online tissue tracking.

RE0: Recognize Everything with 3D Zero-shot Instance Segmentation

2025

Xiaohan Yan*, Zijian Jiang*, Yinghao Shuai*, Nan Wang, Xiaowei Song,

Wenbo Ji, Ge Wu, Jinyu He, Gang Wei, Zhicheng Wang†

ICRA 2025

Training-free 3D zero-shot instance segmentation from multi-view masks and CLIP semantics.

Research Experience

Video-Action World Models for Robot Dexterous Manipulation

Apr 2026 – Present

Agile Robots SE, World Models & Dexterous Manipulation

Research Intern

Mentors: Mahdi Mustapha Hamad

- Developing a cross-embodiment video generation method that translates egocentric human demonstrations into robot-domain videos for downstream policy learning.

Human Motion Video Diffusion

Mar 2026 – Present

TUM Visual Computing Group

Master's Thesis

Advisor: Prof. Matthias Nießner; Co-advisors: Yu Chi, Jiapeng Tang

- Developing camera-controlled human motion video diffusion for controllable synthesis of human motion and scene interactions across changing viewpoints.

Video Diffusion Shader (Head Avatar)

Apr 2025 – Mar 2026

TUM Visual Computing Group

Research Intern

Mentors: Jiapeng Tang, Prof. Matthias Nießner

- Led the development of ViDS, an identity-preserving video diffusion method for long-form portrait animation.
- Ranked first on 8 of 13 reported VFHQ metrics for reenactment quality and identity preservation.

LiteTracker: Online Tissue Tracking

Aug 2024 – Apr 2025

ImFusion | TUM CAMP

Research Intern

Mentors: Mert Asim Karaoglu, Prof. Benjamin Busam, Prof. Nassir Navab, Alexander Ladikos

- Implemented online inference and EMA-flow initialization for LiteTracker (MICCAI 2025), and conducted low-latency tracking experiments.
- Remained competitive on STIR and SuPer while running $\sim 7\times$ faster than its predecessor and $2\times$ faster than prior state of the art, reaching 29.67 ms P95 for 1,024 points.

CSG-Fusion: Sparse-View Gaussian Splatting

Feb 2024 – Aug 2025

TUM CVG | Oxford VGG

Guided Research

Mentors: Prof. Daniel Cremers, Prof. Yan Xia, Weirong Chen, Prof. Chuanxia Zheng

- Proposed and developed CSG-Fusion end-to-end (method idea and all experiments): matching-based pruning, mask-based feature aggregation, and cross-view supervision.
- On ScanNet++, +2.8 dB PSNR over Splatt3R at 90% overlap with $\sim 124K$ fewer Gaussians; also evaluated zero-shot on DTU.

Large-Scale 3D Scene Reconstruction

July 2023 – Sept 2023

Zhejiang University

Research Assistant

Advisor: Prof. Yiyi Liao

- Contributed to the development and experimental evaluation of a large-scale 3D scene reconstruction pipeline.

Endoscopic Scene Reconstruction with 4D Half Gaussian Splatting

2025

Tongji University

Master's Thesis

- Developed a 4D Half-Gaussian splatting pipeline for deformable stereo endoscopic reconstruction with depth-prior initialization, HexPlane spatio-temporal deformation, and edge-aware depth regularization.
- Achieved 38.1 PSNR on EndoNeRF versus prior endoscopic GS/NeRF baselines (e.g., EndoGaussian, Deform3D-GS); also evaluated on SCARED.

Honors

DAAD Scholarship

2023 – 2024

National Encouragement Scholarship (China)

2019